

KHAMIS MUSHAIT, Saudi Arabia

WMO: 411140

Lat: 18.297N

Lon: 42.804E

Elev: 2066

StdP: 78.85

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.5	(c) 8.8	(d) -8.1	(e) 2.4	(f) 18.3	(g) -5.9	(h) 3.0	(i) 19.8	(j) 10.4	(k) 17.8	(l) 9.7	(m) 18.0	(n) 0.8	(o) 180	(p) 0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.3	(c) 32.6	(d) 14.7	(e) 31.9	(f) 15.0	(g) 31.1	(h) 15.0	(i) 19.7	(j) 24.9	(k) 19.1	(l) 24.2	(m) 18.5	(n) 23.8	(o) 5.6	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	16.9	22.3	17.7	16.4	22.1	16.9	15.6	21.7	66.9	25.3	64.5	23.8	62.5	23.7	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.6	(b) 8.6	(c) 7.6	(e) 4.6	(f) 34.2	(g) 1.5	(h) 1.6	(i) 3.5	(j) 35.4	(k) 2.7	(l) 36.3	(m) 1.8	(n) 37.2	(o) 0.7	(p) 38.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			4.6	34.2	1.5	1.6	3.5	35.4	2.7	36.3	1.8	37.2	0.7	38.4		
			0.5	21.5	1.8	1.3	-0.7	22.4	-1.8	23.1	-2.8	23.9	-4.1	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	15.2	17.0	18.6	20.6	23.3	25.1	25.2	24.8	23.7	20.4	17.6	15.7
	DBStd	3.90	1.68	1.69	1.76	1.74	1.70	1.12	1.39	1.19	1.23	1.33	1.53	1.44
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	277	97	44	18	3	1	0	0	0	0	1	31	82
	CDD10.0	3874	162	195	266	319	412	454	471	459	410	322	229	176
	CDD18.3	1110	1	6	26	72	154	204	213	201	160	64	9	1
	CDH23.3	9249	28	67	166	467	1239	1911	1791	1637	1379	452	89	23
CDH26.7	3099	1	2	8	68	405	786	701	647	440	41	2	0	
Wind	WSAvg	3.1	3.1	3.4	3.8	3.3	2.9	3.2	3.2	3.2	3.4	3.1	2.4	2.5
Precipitation	PrecAvg	179	7	6	22	41	30	9	21	26	5	4	5	1
	PrecMax	417	39	57	249	193	78	81	71	140	40	37	31	13
	PrecMin	26	0	0	0	0	0	1	1	3	0	0	0	0
	PrecStd	93	9	12	48	49	24	17	17	29	8	10	8	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	26.8	27.8	29.9	32.1	33.9	34.1	33.1	32.2	28.9	26.8	25.2
		MCWB	12.3	13.7	14.3	14.5	15.1	13.9	14.5	15.0	14.2	12.9	13.2	11.9
	2%	DB	24.0	25.2	26.2	28.2	31.1	32.2	32.4	32.0	31.1	27.9	25.2	23.9
		MCWB	11.6	13.3	13.8	14.7	14.9	14.6	14.9	15.4	14.6	12.9	13.1	12.3
	5%	DB	22.2	24.0	25.1	27.2	30.1	31.5	31.2	31.1	30.1	26.9	24.2	22.8
		MCWB	11.8	12.9	13.8	14.7	14.8	14.7	15.2	15.5	14.4	12.6	13.0	12.4
	10%	DB	20.9	22.8	24.0	26.1	29.1	30.8	30.2	30.2	29.1	26.0	23.1	21.2
		MCWB	11.8	12.9	13.9	14.8	14.9	14.7	15.2	15.4	14.1	12.6	12.9	12.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	17.1	17.6	18.6	19.4	20.0	20.5	20.5	19.3	17.2	17.0	16.2
		MCDB	20.3	23.6	23.2	25.0	24.8	25.4	25.9	26.1	23.9	21.3	22.0	20.9
	2%	WB	14.4	15.7	16.5	17.4	18.5	19.0	19.5	19.7	18.4	16.3	16.1	15.2
		MCDB	19.3	21.5	22.4	23.7	24.2	24.8	24.6	24.7	23.1	20.7	20.6	19.9
	5%	WB	13.8	14.9	15.8	16.8	17.8	18.2	18.7	19.1	17.6	15.7	15.5	14.5
		MCDB	18.5	20.5	21.6	23.1	23.9	24.3	24.1	23.9	22.8	20.6	20.0	19.2
	10%	WB	13.2	14.2	15.2	16.1	17.0	17.3	18.1	18.5	16.7	14.9	14.8	13.9
		MCDB	18.0	19.6	21.0	22.6	23.8	24.3	23.7	23.3	23.1	20.7	19.5	18.5
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.6	10.8	11.3	12.3	12.3	11.6	12.2	12.2	12.1	12.1	12.3
		MCDBR	14.5	13.8	12.6	12.5	13.0	13.0	12.4	12.8	12.9	12.7	13.2	14.4
		MCWBR	7.5	7.0	6.3	5.7	5.5	5.8	5.4	5.5	5.9	6.4	6.6	7.2
	5% WB	MCDBR	11.6	11.9	10.8	11.4	12.5	12.5	11.9	12.3	12.5	12.2	11.8	11.9
		MCWBR	6.4	6.7	5.8	5.6	5.8	6.3	5.7	5.8	6.6	7.0	6.2	6.5
Clear-Sky Solar Irradiance	taub	0.327	0.365	0.403	0.450	0.508	0.564	0.624	0.610	0.502	0.390	0.369	0.333	
	taud	2.412	2.265	2.144	2.019	1.844	1.690	1.611	1.635	1.837	2.157	2.248	2.378	
	Ebn at Noon	958	933	906	862	803	752	709	724	804	895	901	938	
	Edh at Noon	110	134	157	179	211	244	264	260	210	147	128	111	
All-Sky Solar Radiation	RadAvg	5.12	5.87	6.60	6.88	7.19	7.25	6.59	6.36	6.58	6.38	5.41	5.01	
	RadStd	0.26	0.29	0.31	0.39	0.37	0.25	0.28	0.21	0.20	0.22	0.19	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	-34	+129	+99
(47) Regional (4 neighbors)		+0.47	N/A	N/A	+0.49	N/A	N/A	N/A	-11	+192	+154

Nomenclature: See separate page